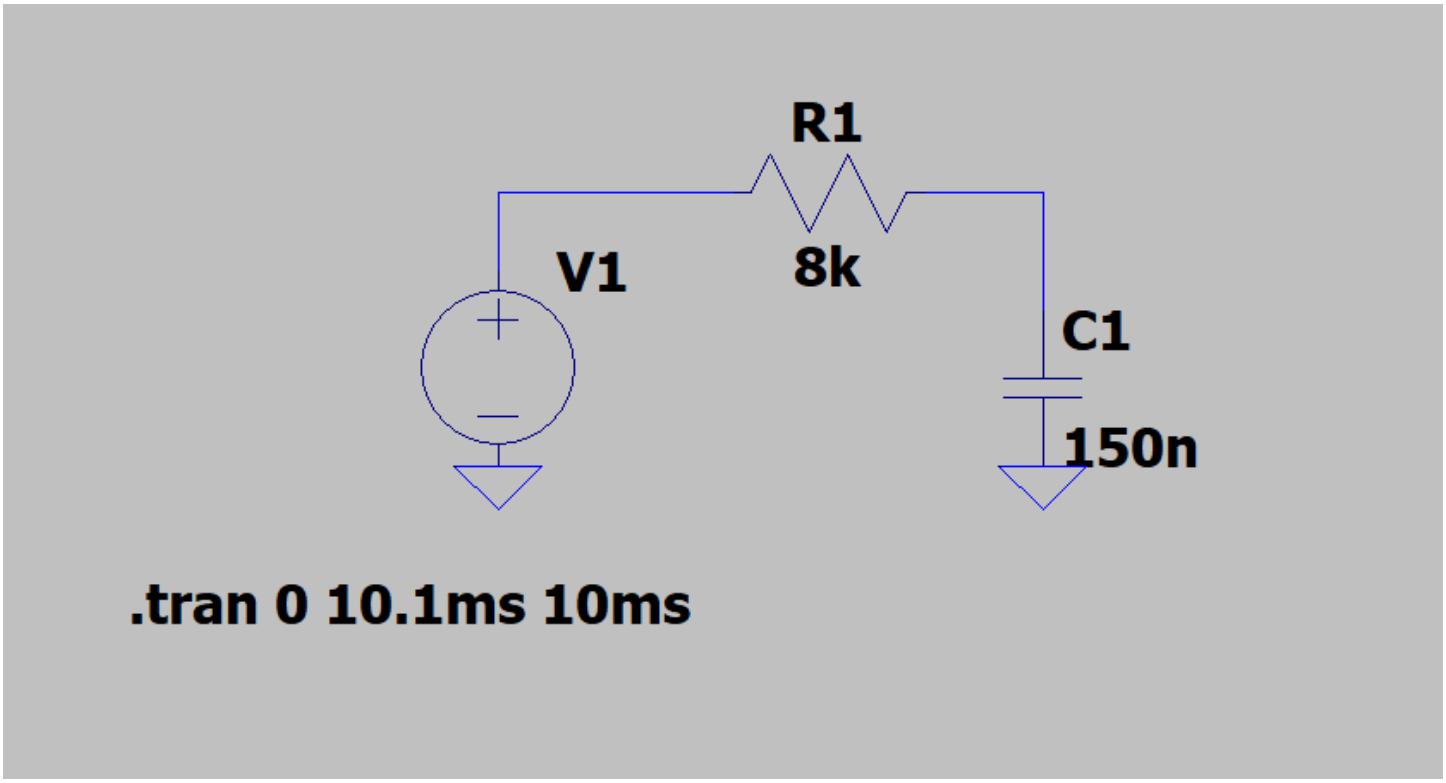
Ain Shams University
Faculty of Engineering
Electronics & Comm. Eng. Dept.



Experiment (1) Report

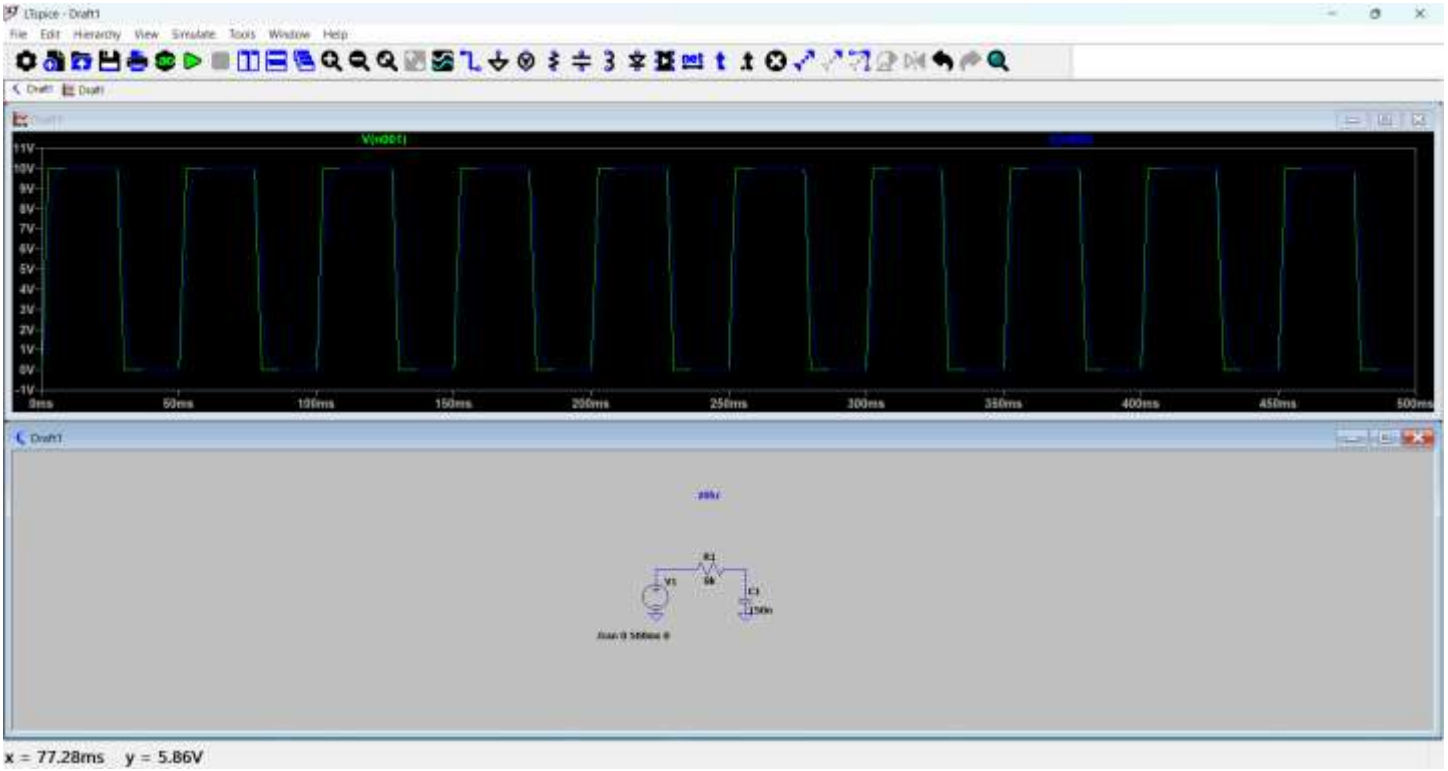
Names	IDs	Department	Attendance slot	Group No.	Group Code Part A, B
Abdelrahman sherif	2300100	ECE	even	29	A8
Amr hatem	2300729	ECE			
Abdelrahman Mohamed ali	2300518	ECE			

1. Construct the circuit

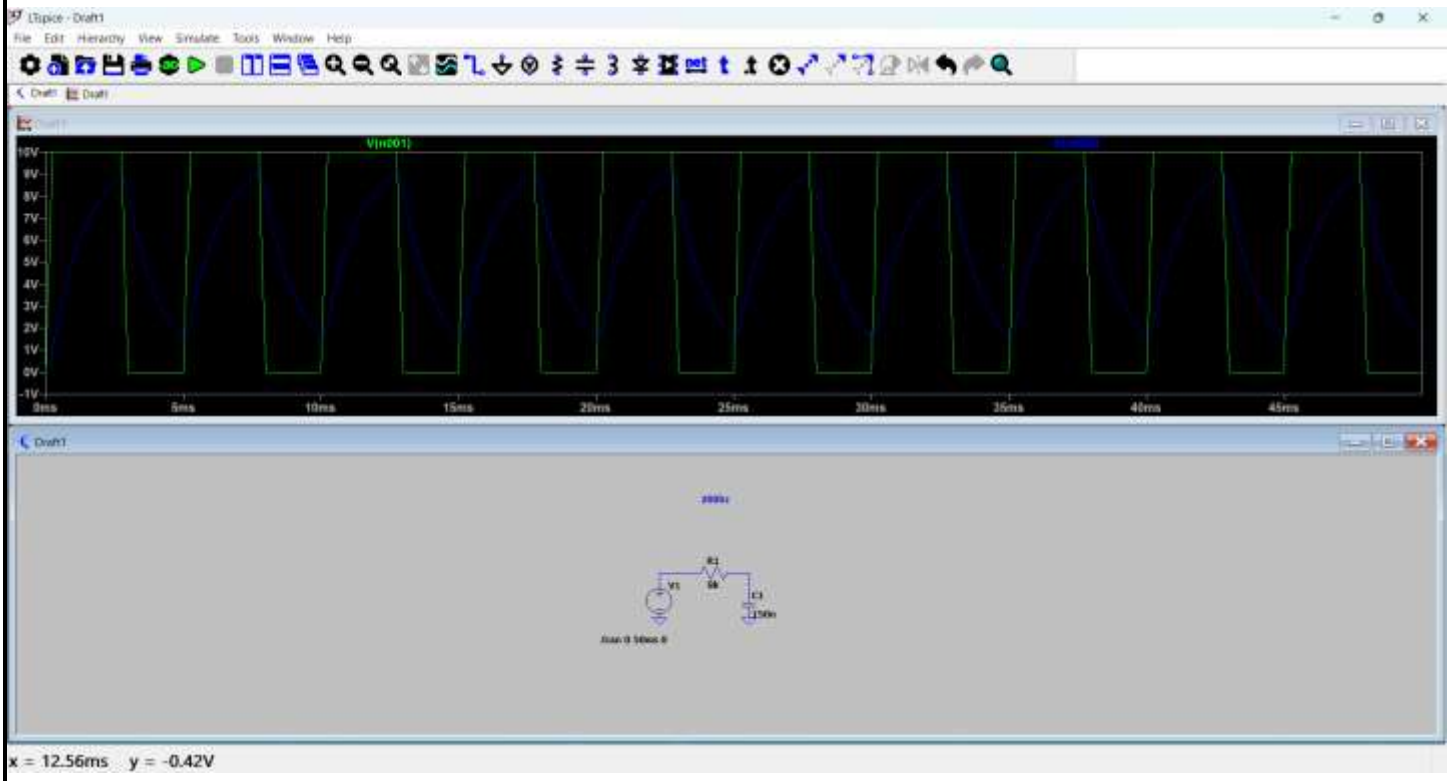


2.change the source freq

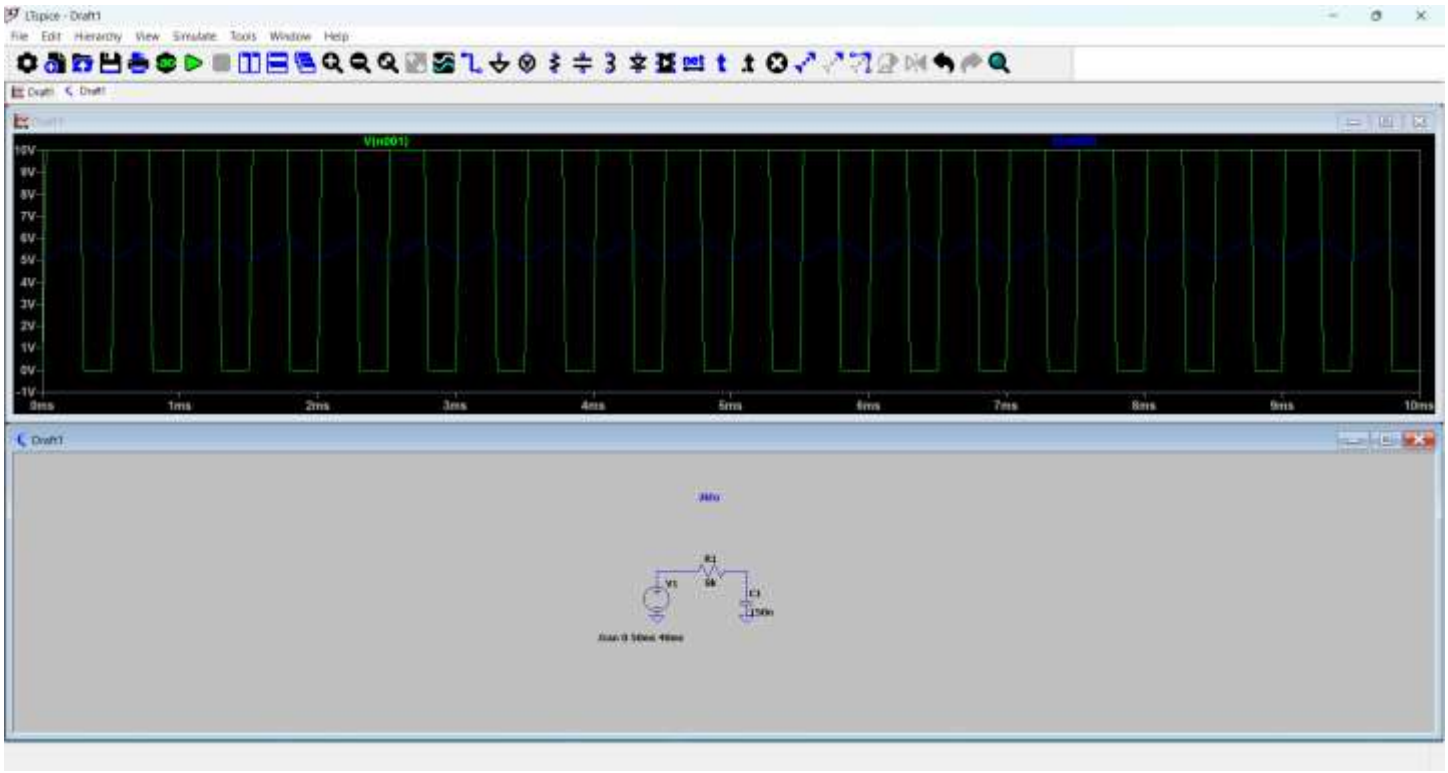
20hz



200hz

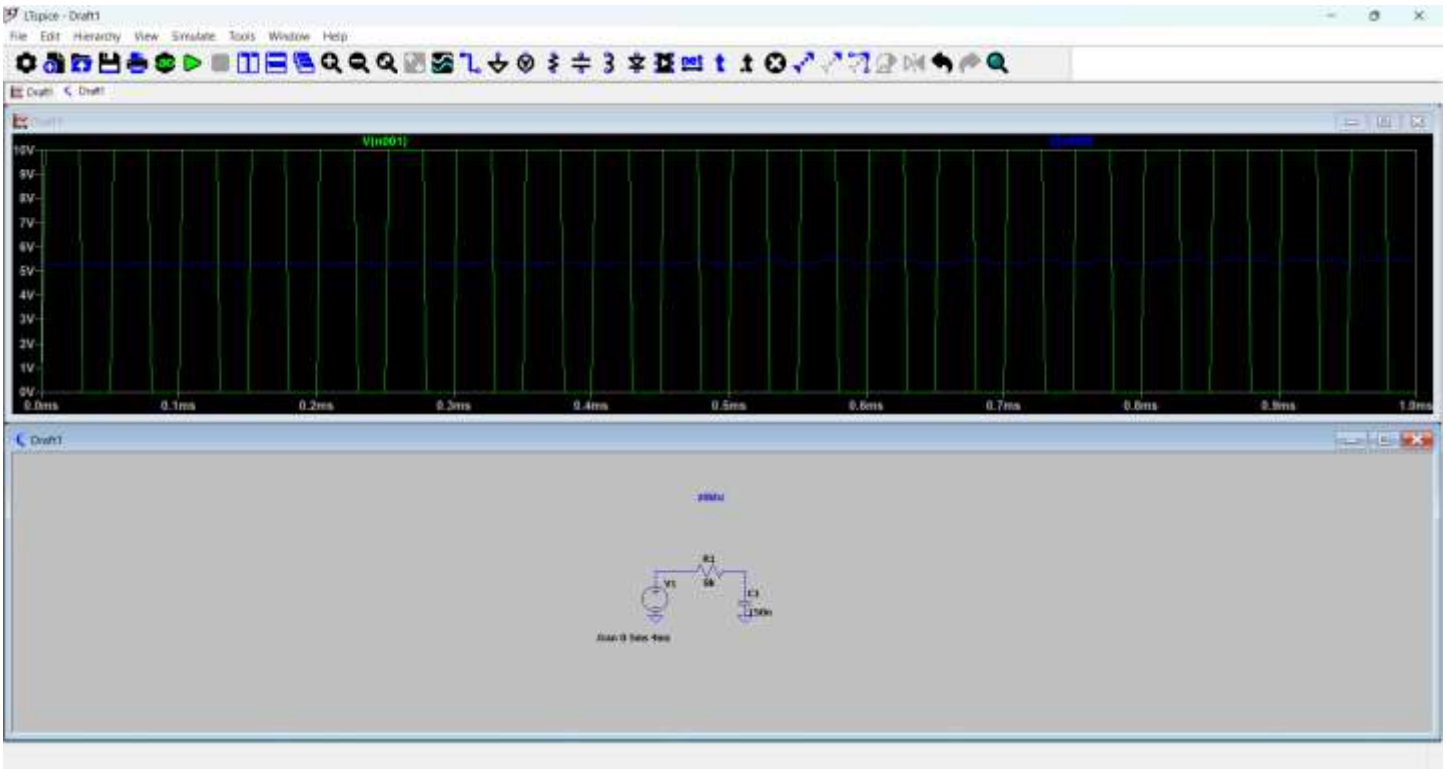

LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
Draft1
V(in001)
10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
-1V
0ms
5ms
10ms
15ms
20ms
25ms
30ms
35ms
40ms
45ms
Draft1
R1
V1
C1
x = 12.56ms y = -0.42V

2khz

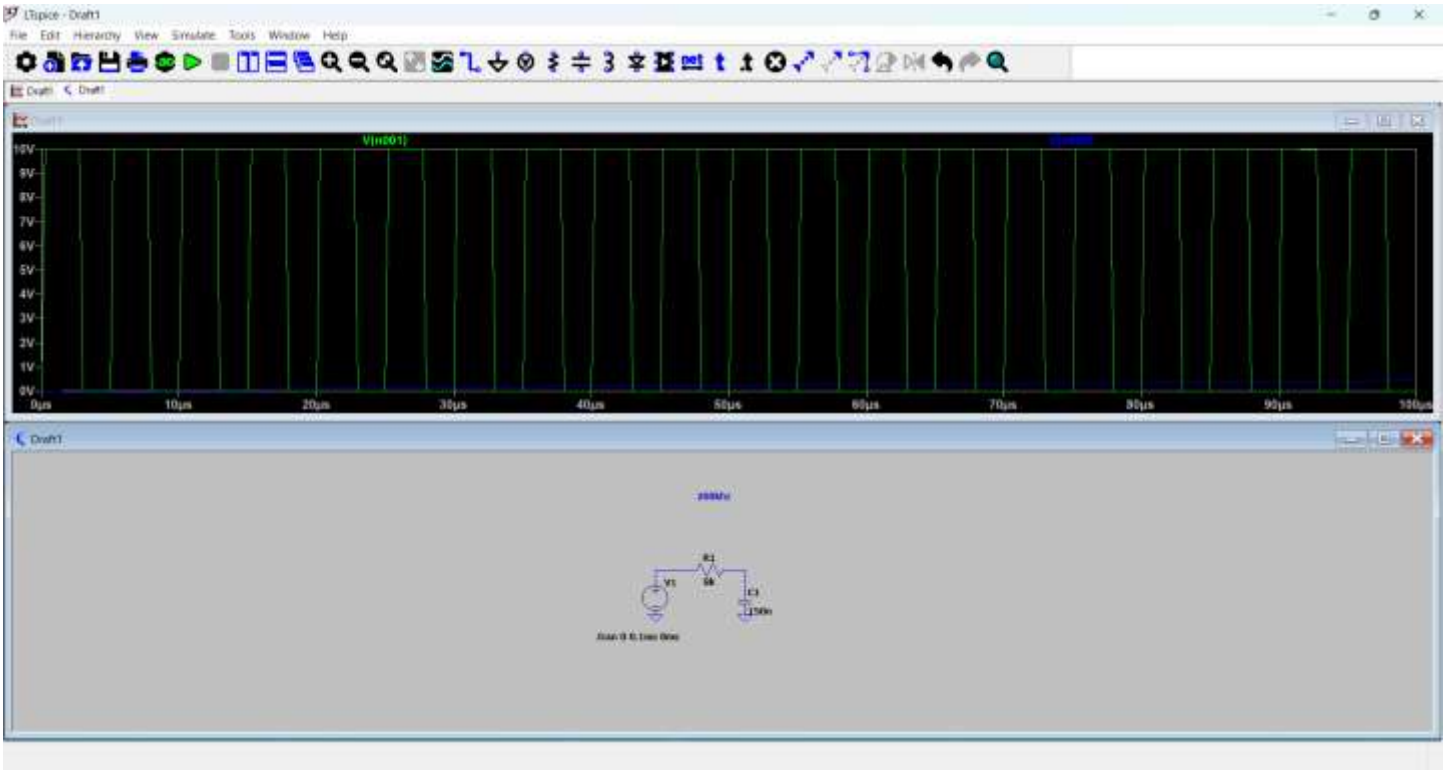

LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
V(in001)
10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
-1V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
Draft1
R1
V1
C1

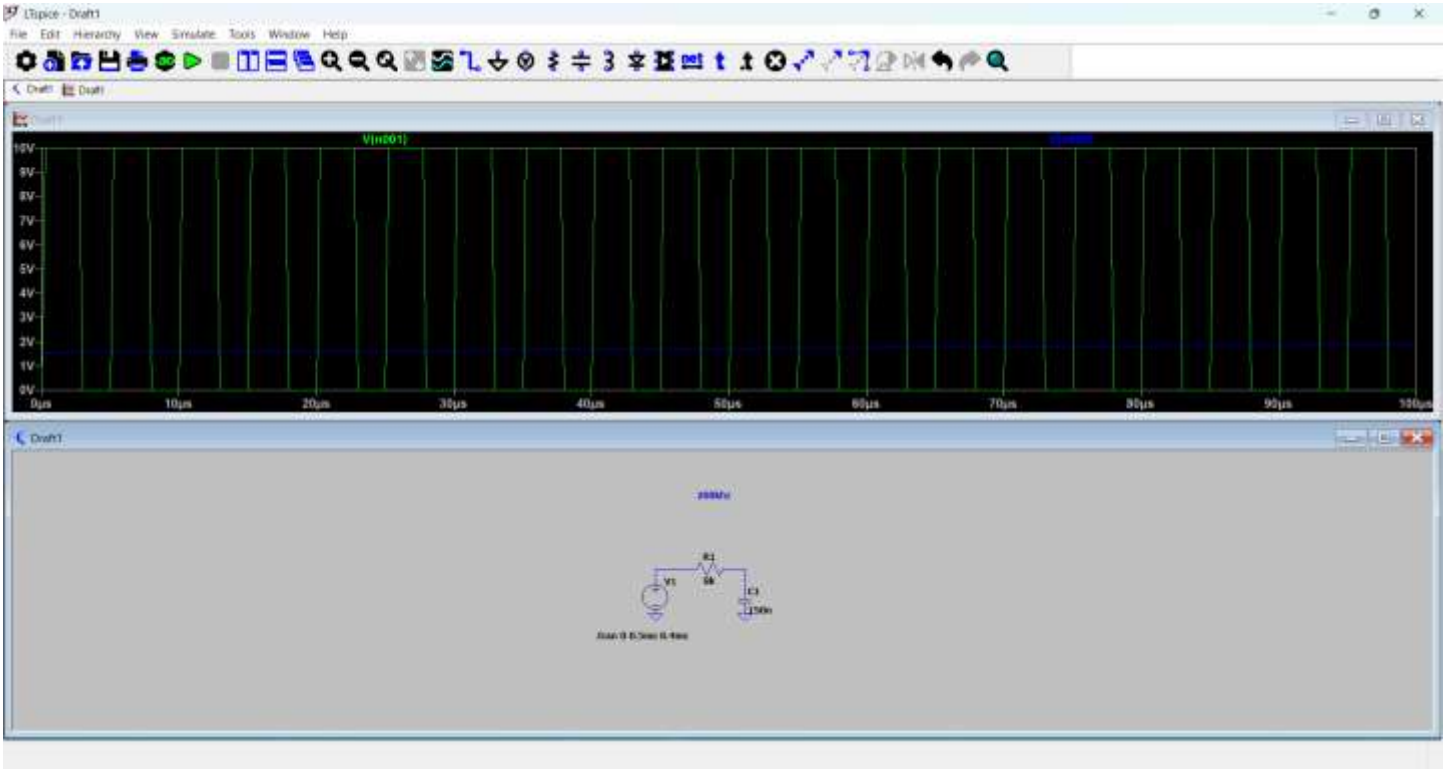
20khz


LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
V(in001)
11V
10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
0.0ms
0.1ms
0.2ms
0.3ms
0.4ms
0.5ms
0.6ms
0.7ms
0.8ms
0.9ms
1.0ms
Draft1
20kHz
R1
5k
V1
C1
150n

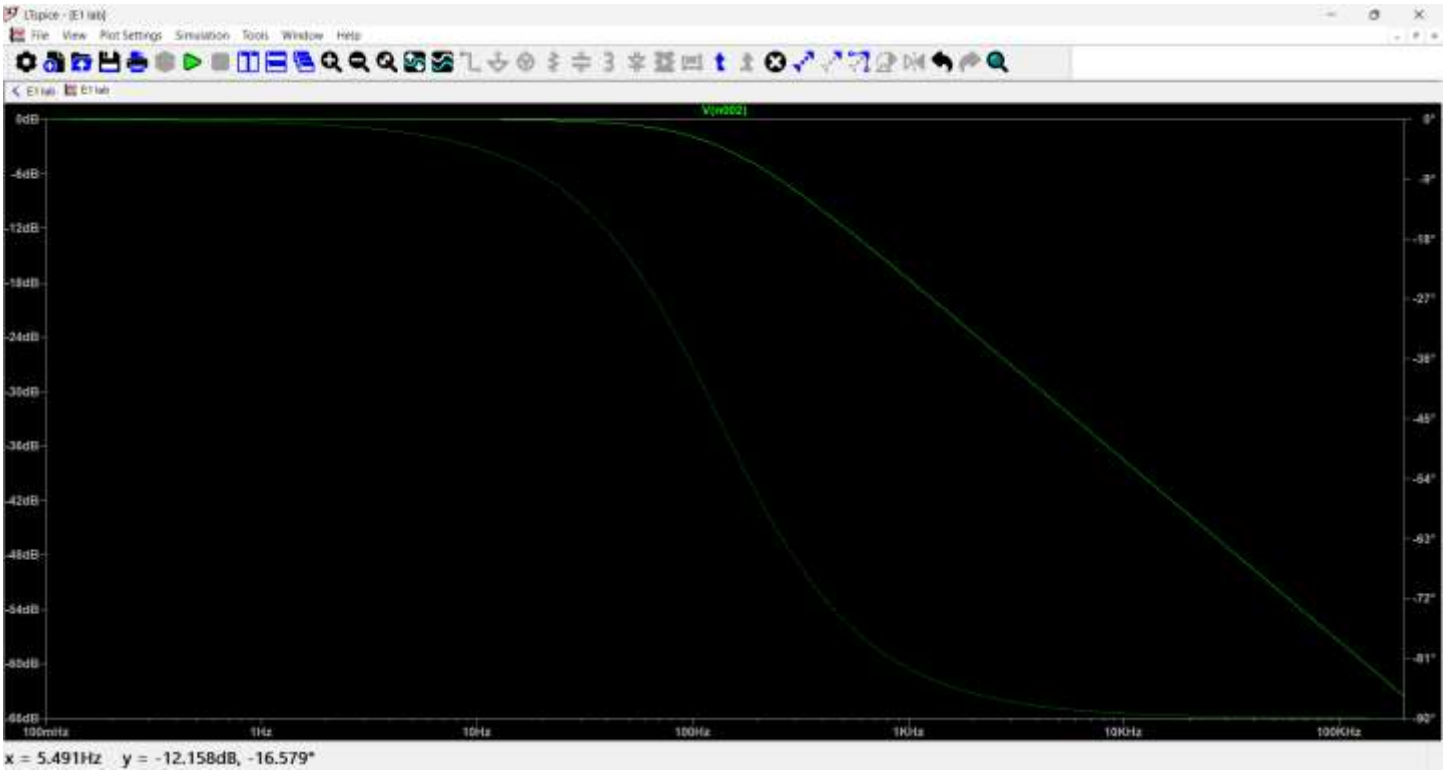

LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
V(in001)
10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
0.0ms
0.1ms
0.2ms
0.3ms
0.4ms
0.5ms
0.6ms
0.7ms
0.8ms
0.9ms
1.0ms
Draft1
20kHz
R1
5k
V1
C1
150n

200khz


LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
V(in001)
10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
0µs
10µs
20µs
30µs
40µs
50µs
60µs
70µs
80µs
90µs
100µs
Draft1
200khz
R1
6k
V1
C1
150n


LTspice - Draft1
File Edit Hierarchy View Simulate Tools Window Help
V(in001)
10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
0µs
10µs
20µs
30µs
40µs
50µs
60µs
70µs
80µs
90µs
100µs
Draft1
200khz
R1
6k
V1
C1
150n

3.draw the frequency response



4.what happens to the square wave shape while increasing frequency?

the square wave edges are curved more and not sharp and it changes to a sinc